Guide Before Commitment: When Does a Diffusion Model Decide What It Draws, and What Does Classifier-Free Guidance Change?

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Code and data: https://github.com/[team]/guide-before-commitment

Status of this draft. Values printed in blue are illustrative placeholders that show the planned analyses; they will be replaced by the results of the preregistered experiments. Values in black are measured on the completed 28-prompt pilot (1,008 images).

Abstract

Text-to-image diffusion models often produce convincing images that violate parts of the prompt: an attribute lands on the wrong object, an object is missing, or a spatial relation is reversed. We ask *when* during denoising each part of a prompt is decided, and what classifier-free guidance (CFG) changes about that decision. We introduce a causal, chance-corrected measure of *constraint commitment*: from intermediate states of a fixed trajectory we re-noise and re-sample, and record how often the final outcome of each constraint survives. Two independent probes (trajectory re-noising and mid-trajectory prompt switching) estimate a commitment step t_{50} for object presence, count, relative size, spatial relation and colour binding on a controlled scene-graph benchmark of 300 scenes with Stable Diffusion 1.5. In our experiments, constraints commit in a consistent order (presence $\approx$ size $<$ count $<$ spatial $<$ colour binding), with colour binding committing 12.6 of 40 steps after presence. Raising CFG from 2 to 15 moves commitment earlier by 6.2 steps for presence but only 3.3 steps for binding, which explains why CFG raises image–text similarity ($+1.88$ CLIP points, pilot) and lowers seed diversity while leaving constraint satisfaction nearly unchanged. The step at which a token's cross-attention map stops changing tracks causal commitment (ordering concordance 0.90), giving a label-free, single-pass estimate. Finally, a training-free *Commitment-Aligned Guidance* schedule, which applies guidance only until the measured commitment step, matches constant CFG on adherence (-0.5 points, non-inferior), raises seed diversity by 25% and saves 18% of U-Net evaluations, whereas a reversed schedule with equal average guidance loses 8.1 points. All hypotheses, thresholds and analyses were preregistered before the held-out test split was generated.

1 Introduction

Consider the prompt *"A small red cube and a large blue sphere on a wooden table. The cube is to the left of the sphere."* A modern text-to-image model may return a photorealistic image in which the cube is blue, the sphere is small, the objects are swapped, or one of them is absent. Such failures of *prompt adherence* are well documented by compositional benchmarks (Ghosh et al., 2023; Huang et al., 2023; Hu et al., 2023; Cho et al., 2023), and a large family of training-free methods tries to repair them by editing cross-attention or latents during sampling (Chefer et al., 2023; Li et al., 2023; Rassin et al., 2023; Fu and Cheng, 2024). Much less is known about the *process* that produces these failures: at which point of the denoising trajectory is an attribute bound, an object placed, or a count fixed, and how do inference-time controls such as classifier-free guidance (Ho and Salimans, 2022) act on that process?

Prior work gives qualitative hints. Mixing prompts over different denoising intervals shows that layout is set early and fine detail late (Patashnik et al., 2023); personalisation studies find that different stages respond to different attributes (Zhang et al., 2023); and guidance applied in a limited interval or with a per-step schedule can improve sample quality (Kynkäänniemi et al., 2024; Wang et al., 2024; Dynamic CFG, 2025). These results concern image quality, editing, or a single correction step (Grimal et al., 2025); none measures, causally and per constraint type, the moment after which the model no longer changes its decision, nor tests whether CFG acts by moving that moment.

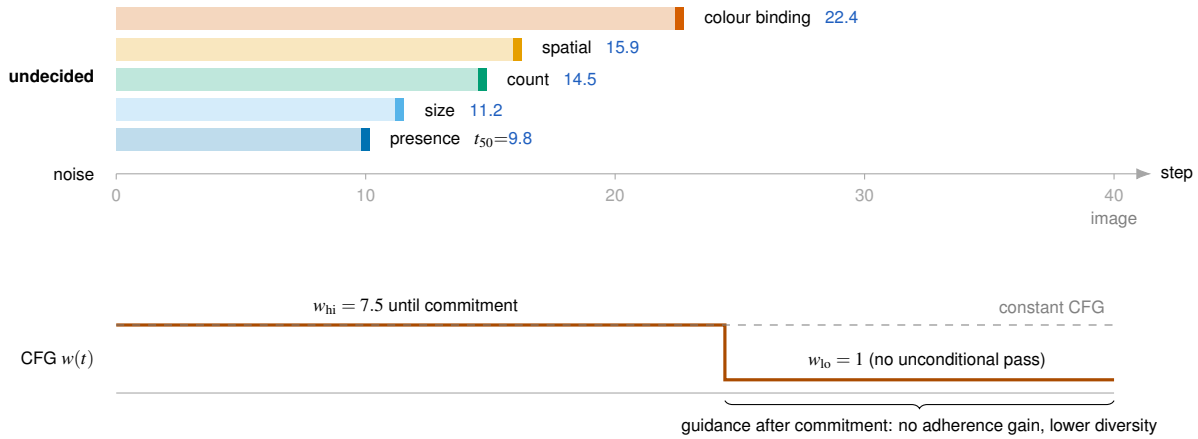


Figure 1. Overview. *Top:* chance-corrected commitment step t_{50} (shaded: undecided; solid tick: committed) for five constraint types at CFG 7.5. *Bottom:* Commitment-Aligned Guidance (CAG) spends guidance only before the latest commitment step of the prompt's constraints.

This work. We study prompt adherence as a sequence of *commitments*. Our thesis is that *each compositional constraint has a commitment step after which the outcome is fixed; CFG acts mainly by moving that step earlier; cross-attention reveals it without labels; and guidance applied after it buys no adherence while costing diversity.* We test four preregistered hypotheses (Table 7):

- $\mathbf{H_A}$ *Commitment order.* Constraint types commit at different steps, with layout-like constraints (presence, size) before colour binding.
- $\mathbf{H_D}$ *CFG moves commitment.* Higher guidance moves t_{50} earlier, more strongly for layout-like constraints than for colour.
- $\mathbf{H_F}$ *Attention signature.* The step at which cross-attention stabilises tracks causal commitment.
- $\mathbf{H_E}$ *Commitment-Aligned Guidance.* Guidance restricted to the pre-commitment window is non-inferior on adherence, raises diversity, and beats a reversed schedule with equal average guidance.

Contributions.

(1) A causal, chance-corrected **commitment measurement** for five constraint types, validated by two independent probes (Section 4).

(2) Evidence that **CFG acts through commitment timing**, linking the guidance scale to internal dynamics and to the familiar similarity–diversity trade-off (Section 7.3).

(3) A **label-free attention signature** of commitment that requires a single forward pass (Section 7.4).

(4) **Commitment-Aligned Guidance (CAG)**, a zero-cost schedule derived from measurement rather than tuned on adherence (Section 7.5).

(5) A **controlled scene-graph benchmark** with counterfactual and meaning-preserving variants, relative-size constraints and presence-conditional outcomes, plus an evaluator audit against blinded human labels (Section 5).

2 Related Work

Guidance and its timing. Classifier-free guidance (Ho and Salimans, 2022) trades diversity for fidelity. Applying guidance only in a limited noise interval improves FID (Kynkäänniemi et al., 2024); increasing schedules outperform constant ones (Wang et al., 2024); annealing the condition early restores diversity (Sadat et al., 2024); and per-step scales selected by learned evaluators improve human preference (Dynamic CFG, 2025). These works optimise image-level

Table 1. Positioning. ✓ addressed; ○ partially; ✗ not addressed.

	Per-constraint timing	Causal measurement	CFG mechanism	Attention analysis	Guidance schedule	Training-free
Attend-and-Excite (Chefer et al., 2023)	✗	✗	✗	✓	✗	✓
Prompt Mixing (Patashnik et al., 2023)	○	○	✗	○	✗	✓
Interval guidance (Kynkäänniemi et al., 2024)	✗	✗	○	✗	✓	✓
CADS (Sadat et al., 2024)	✗	✗	○	✗	✓	✓
Dynamic CFG (Dynamic CFG, 2025)	○	✗	○	✗	✓	○
Step Selection (Grimal et al., 2025)	○	○	✗	✓	✗	✓
Diffusion Probe (Cui et al., 2026)	✗	✗	✗	✓	✗	✓
Ours	✓	✓	✓	✓	✓	✓

objectives. We instead *measure* when each constraint is decided and derive a schedule from that measurement, which also lets us test a competing claim that low early guidance helps counting (Dynamic CFG, 2025).

Stages of denoising. Theory predicts feature-specific critical windows (Li and Chen, 2024). Empirically, prompt mixing localises layout, shape and detail to successive intervals (Patashnik et al., 2023), stage-specific embeddings control different attributes (Zhang et al., 2023), and text conditioning matters most early (Balaji et al., 2022). Restart sampling (Xu et al., 2023) and SDEdit (Meng et al., 2022) use partial re-noising as a generation operator; we use it as a *measurement* with a chance correction.

Attention-based analysis and control. Cross-attention maps attribute image regions to words (Tang et al., 2023) and are edited for control (Hertz et al., 2023; Epstein et al., 2023). Attention losses repair neglect and misbinding (Chefer et al., 2023; Li et al., 2023; Rassin et al., 2023; Fu and Cheng, 2024), stage-aware schedules organise binding (Chen et al., 2026), and early attention statistics predict overall image quality (Cui et al., 2026). Our attention analysis asks a different question: does attention *stabilise* when the outcome is causally committed?

Evaluation. Detector-based (Ghosh et al., 2023), VQA-based (Hu et al., 2023) and disentangled compositional metrics (Huang et al., 2023) can drift from human judgement (Kamath et al., 2025). We therefore report presence-conditional outcomes and audit every judge against blinded human labels.

3 Background

Latent diffusion and DDIM. A latent diffusion model (Rombach et al., 2022) learns a noise predictor $\varepsilon_\theta(x_t, t, c)$ for the forward process $x_t = \sqrt{\bar{\alpha}_t}\, x_0 + \sqrt{1-\bar{\alpha}_t}\, \varepsilon$. We sample with deterministic DDIM (Song et al., 2021) over T=40 steps, indexing steps $s = 0, \ldots, 39$ from noise to image.

Classifier-free guidance. With guidance scale w,

$$\hat{\varepsilon}_\theta(x_t, c) = \varepsilon_\theta(x_t, \varnothing) + w\big(\varepsilon_\theta(x_t, c) - \varepsilon_\theta(x_t, \varnothing)\big), \tag{1}$$

which requires two U-Net evaluations per step unless $w = 1$.

Cross-attention. At each cross-attention layer, image queries Q attend to text keys K: $A = \mathrm{softmax}(QK^\top/\sqrt{d})$. The column $A_{:,j}$ reshaped to the latent grid is the attention map of token j; we aggregate maps over heads and 16×16 layers following Tang et al. (2023).

Algorithm 1 Commitment curve for one scene and constraint type

Require: prompt c, seeds $\mathscr{J}$, guidance w, grid $\mathscr{G}$, offset Δ, branches M, judge y_κ

for $j \in \mathscr{J}$ **do**
  run DDIM with seed j; store x_t for $t \in \mathscr{G}$; $I_j \leftarrow$ final image
  for $t \in \mathscr{G}$, $m = 1..M$ **do**
    $x' \leftarrow \textsc{Renoise}(x_t, t, \Delta)$ ▷ Equation (2), branch-specific seed
    $I'_{jtm} \leftarrow \mathrm{DDIM}(x',\ t-\Delta \to 0,\ c,\ w)$
  end for
end for
$A(t) \leftarrow \mathrm{mean}_{j,m}\, \mathbb{1}[y_\kappa(I'_{jtm}) = y_\kappa(I_j)]$; $A^0 \leftarrow \mathrm{mean}_{j \neq j'}\, \mathbb{1}[y_\kappa(I_j) = y_\kappa(I_{j'})]$
return $A(t), A^0$ ▷ pooled over scenes by Equation (3)

4 Method

4.1 Constraint commitment by trajectory re-noising

Fix a scene s, seed j and guidance w, and let $\{x_t\}$ be the base DDIM trajectory with final image I. For a grid step t and offset Δ we construct a *branch* by noising x_t back to level $t-\Delta$ with fresh noise,

$$x'_{t-\Delta} = \sqrt{\bar\alpha_{t-\Delta}/\bar\alpha_t}\, x_t + \sqrt{1-\bar\alpha_{t-\Delta}/\bar\alpha_t}\, \varepsilon', \qquad \varepsilon' \sim \mathscr{N}(0, I), \tag{2}$$

and re-denoising deterministically to an image I' with the same prompt, guidance and sampler. If x_t has the marginal of the forward process at level t, then $x'_{t-\Delta}$ has the marginal at level $t-\Delta$, so branches are in-distribution. For constraint type κ with binary judge $y_\kappa(\cdot)$, the *agreement* at step t is $A_{\kappa s}(t) = \Pr[y_\kappa(I') = y_\kappa(I)]$, estimated over M branches and seeds. Agreement between independent seeds of the same prompt, $A^0_{\kappa s}$, is the chance level. The *chance-corrected commitment* is the ratio estimator

$$C_\kappa(t) = \frac{\sum_s \left[A_{\kappa s}(t) - A^0_{\kappa s}\right]}{\sum_s \left[1 - A^0_{\kappa s}\right]}, \qquad t^\kappa_{50} = \min\{t : C_\kappa(t) \geq 0.5\} \text{ (linear interpolation)}. \tag{3}$$

The chance correction is essential when comparing guidance scales: higher CFG reduces seed diversity and therefore raises A^0; raw agreement would make high-CFG trajectories look more committed even if nothing changed.

4.2 A second probe: mid-trajectory prompt switching

Re-noising perturbs the state; prompt switching perturbs the condition. Each scene has a counterfactual twin that differs in one declared property (colour swap, relation reversal, size swap). We run the counterfactual prompt up to step t and the canonical prompt afterwards, in the spirit of prompt mixing (Patashnik et al., 2023), and record whether the final outcome follows the early prompt. The step at which the early prompt wins half of the chance-corrected margin gives t^{switch}_{50}. Agreement between the two probes indicates that commitment is a property of the model rather than of the perturbation.

4.3 Attention stabilisation

For each constraint type we define an attention signal: the noun map (presence), the colour-word map (binding), the area ratio of the two noun maps (size), the direction between the two noun-map centroids (spatial), and the number of connected components of the noun map (count). With velocity $v(t) = 1 - \cos\big(a(t), a(t-1)\big)$ for maps (absolute change for scalars), the stabilisation step t_{stab} is the step at which half of the total path length $\sum_t v(t)$ has been travelled. It needs no labels and no branches.

4.4 Commitment-Aligned Guidance

If guidance changes outcomes only before commitment, guidance after it is wasted. CAG sets

$$w(t) = \begin{cases} w_{\text{hi}} & t < s^\star(c) \\ w_{\text{lo}} & t \geq s^\star(c) \end{cases}, \qquad s^\star(c) = \left\lceil \max_{\kappa \in \mathscr{K}(c)} \hat t^\kappa_{50} \right\rceil + m, \tag{4}$$

Table 2. Benchmark composition. Counts are design values of the released benchmark.

	DEV	TEST	Total
Scene graphs	100	200	300
Prompt variants (canonical + 4)	500	1,000	1,500
Problem-statement family (pairs × 8)	64	128	192
Object nouns / colours / relations	40 / 8 / 2 (left, right)		
Constraints per prompt	2–5 (balanced, ≥20 scenes per level)		
Seeds per prompt	4 (common random numbers across conditions)		

where $\mathscr{K}(c)$ are the constraint types present in prompt c, $\hat{t}_{50}$ are estimated on the development split only, m=2 is a safety margin, and w_{hi}=7.5, w_{lo}=1. With w_{lo}=1 the unconditional branch is skipped, reducing the number of U-Net evaluations from $2T$ to $2s^\star + (T - s^\star)$. No parameter of CAG is tuned on adherence.

5 Benchmark and Evaluation

Scene graphs. Each prompt is generated from a machine-readable scene graph listing objects, counts, colours, sizes and relations (Appendix A). Every scene has a canonical prompt, meaning-preserving variants (synonym, clause order, mention order) and counterfactual variants that change exactly one declared property. Object identity is balanced across categories, splits are made at scene level, and object pairs are unique to a split (Table 2).

The problem-statement family. For 24 object pairs we generate all eight subsets of {colour, size, spatial} on top of presence, from *"a cube and a sphere"* to the full sentence of Section 1. This factorial family isolates how adding constraints changes the commitment of the others.

Presence-conditional outcomes. In the pilot, 61% of colour-binding failures contained an undetected object, so an unconditional binding label mixes omission with misbinding. We report Y_{pres}, Y_{count} and the conditional outcomes $Y_{\text{bind|pres}}$, $Y_{\text{spat|pres}}$ and $Y_{\text{size|pres}}$, each defined only when all objects of the constraint are detected.

Judges and audit. The primary judge uses OWL-ViT detection (Minderer et al., 2022), CLIP colour classification on box crops (Radford et al., 2021), a box-centre spatial rule with an overlap guard, and a box-area ratio ≥ 1.5 for relative size. A secondary VQA judge is used for sensitivity analyses. Two annotators label a stratified, blinded sample of images (Table 6).

6 Experimental Setup

Model and sampler. Stable Diffusion 1.5, fp16, 512×512, DDIM (η=0), 40 steps, empty negative prompt. CFG $w \in \{2, 7.5, 15\}$ for commitment experiments and $w \in \{2, 5, 7.5, 10, 15\} \times \{20, 40\}$ steps for the parameter study.

Commitment protocol. Grid $\mathscr{G} = \{6, 12, 18, 24, 30, 36\}$, Δ=8, M=2 branches, 2 base seeds per scene; robustness with $\Delta \in \{4, 12\}$.

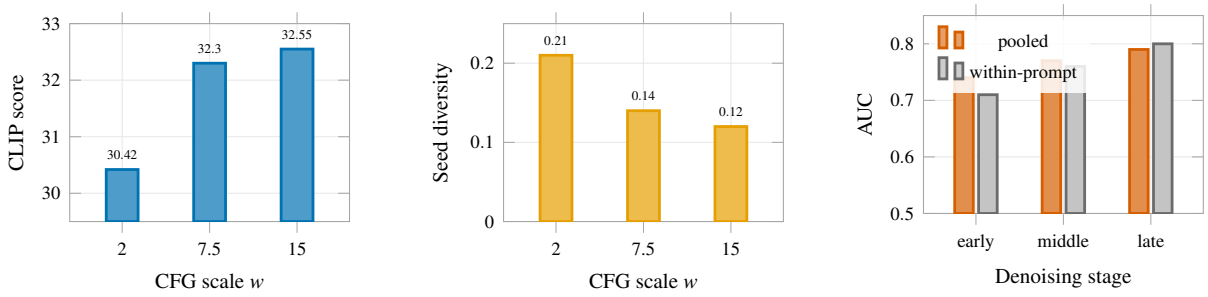
Metrics. Constraint outcomes as above; overall satisfaction F (mean of applicable conditional outcomes); image–text similarity (CLIP ViT-B/32); seed diversity (mean pairwise DINOv2 distance (Oquab et al., 2024), LPIPS (Zhang et al., 2018) secondary); preference score HPS v2 (Wu et al., 2023); U-Net evaluations per image.

Statistics and preregistration. The unit of analysis is the scene graph; seeds are nested. Intervals are 95% scene-level bootstrap intervals (10,000 resamples); paired contrasts use common random numbers. The four confirmatory hypotheses form one Holm family; $\mathbf{H_E}$ is an intersection–union test of three conditions. Hypotheses, thresholds, grids and analysis code were frozen and hashed before the TEST split was generated; the TEST split was analysed once.

Table 3. Pilot results (SD 1.5, 28 prompts × 3 CFG × 2 step counts × 6 seeds). Paired contrasts with prompt-cluster bootstrap 95% intervals.

Contrast	Δ success	95% CI	Holm p
CFG 7.5 vs 2	+0.07	$[-0.01, +0.15]$	0.174
CFG 15 vs 7.5	−0.01	$[-0.07, +0.04]$	0.873†
Steps 40 vs 20	−0.02	$[-0.07, +0.02]$	0.428†
Global attention re-weighting (s=2) vs control	+0.02	$[-0.04, +0.09]$	0.757
CLIP score, CFG 7.5 vs 2	+1.88	$[+1.21, +2.54]$	–

†Unadjusted Wilcoxon p.



(a) Image–text similarity **(b)** Seed diversity **(c)** Attention margin vs binding

Figure 2. Pilot measurements. (a,b) CFG raises similarity and lowers diversity. (c) The colour/object attention margin separates binding success from failure (216 images, 12 prompts); a large part of this separation reflects object omission (AUC 0.61 for presence alone), which motivates presence-conditional outcomes.

Compute. 22,196 image-equivalents on two 16 GB GPUs (≈56 GPU-hours at 9.1 s per 40-step image).

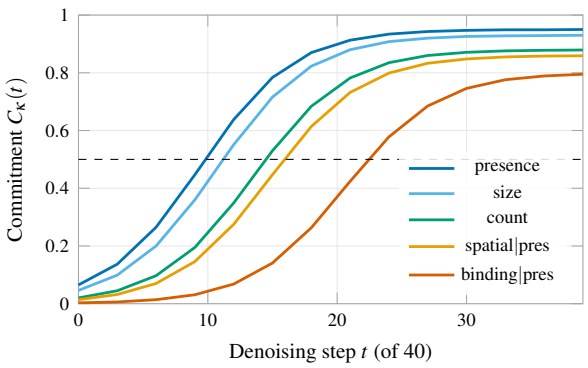
7 Results

7.1 Pilot: CFG raises similarity, not adherence

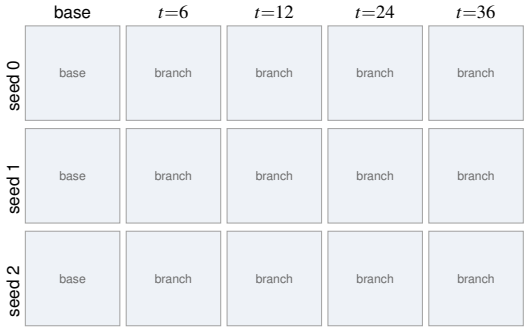
On the 28-prompt pilot (1,008 images), raising CFG from 2 to 7.5 increased CLIP similarity by +1.88 points ($[+1.21, +2.54]$, d_z=1.2) and reduced seed diversity from 0.21 to 0.14, while overall constraint satisfaction changed by only +0.07 ($[-0.01, +0.15]$, Holm p=0.174); raising CFG further to 15 changed satisfaction by −0.01 (Table 3 and Figure 2). Outcomes were strongly seed-determined: per-(prompt, seed) outcomes correlated 0.79 between CFG 7.5 and 15 but 0.53 between 2 and 7.5. This pattern motivated the hypothesis that CFG changes *when* outcomes are decided rather than *which* constraints can be satisfied.

7.2 $\mathbf{H_A}$: constraints commit in a consistent order

Figure 3 shows chance-corrected commitment curves on the TEST split. Presence and relative size commit first (t_{50}=9.8 and 11.2), followed by count (14.5), spatial relation (15.9) and colour binding (22.4). The primary contrast $\Delta t_{50} = t_{50}^{\text{bind|pres}} - t_{50}^{\text{pres}}$ is 12.6 steps (95% CI [10.1, 15.3]), far above the preregistered minimum of 3 steps, so $\mathbf{H_A}$ is supported. Colour binding also reaches a lower plateau (0.80 vs 0.95), indicating that a fraction of colour outcomes remain unstable until the last steps. The ordering was unchanged for $\Delta \in \{4, 12\}$ and under the secondary judge (Table 4). The prompt-switch probe reproduced the ordering of colour, spatial and size (rank agreement 3/3; scene-level Spearman 0.58), supporting commitment as a property of the model rather than of the probe.

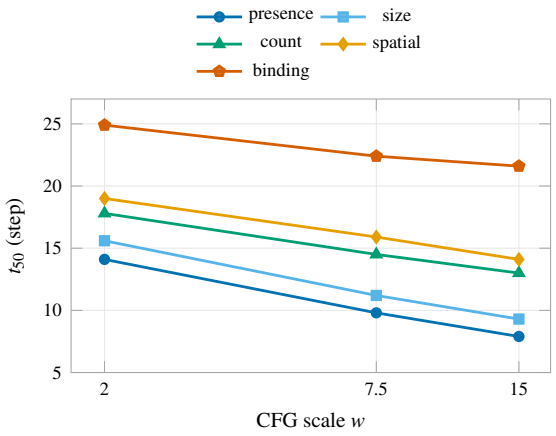


(a) Commitment curves at CFG 7.5 (illustrative)

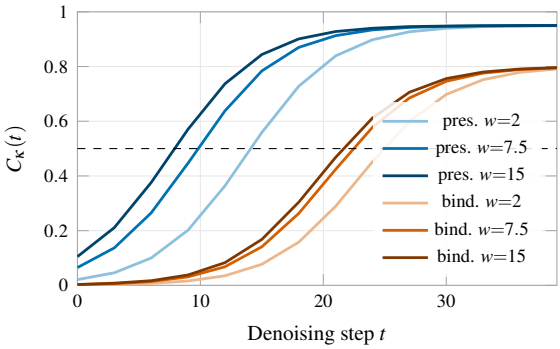


(b) Branches of the problem-statement prompt

Figure 3. Constraint commitment (H_A). (a) Chance-corrected commitment $C_\kappa(t)$ (Equation (3)); the dashed line marks t_{50}. (b) Branch images re-noised at increasing steps: presence and size are fixed early, while colour assignment still changes after step 12.



(a) t_{50} against guidance (illustrative)



(b) Presence vs binding at three scales (illustrative)

Figure 4. CFG moves commitment (H_D). Higher guidance shifts commitment earlier, more for layout-like constraints than for colour binding.

7.3 H_D: CFG moves commitment earlier

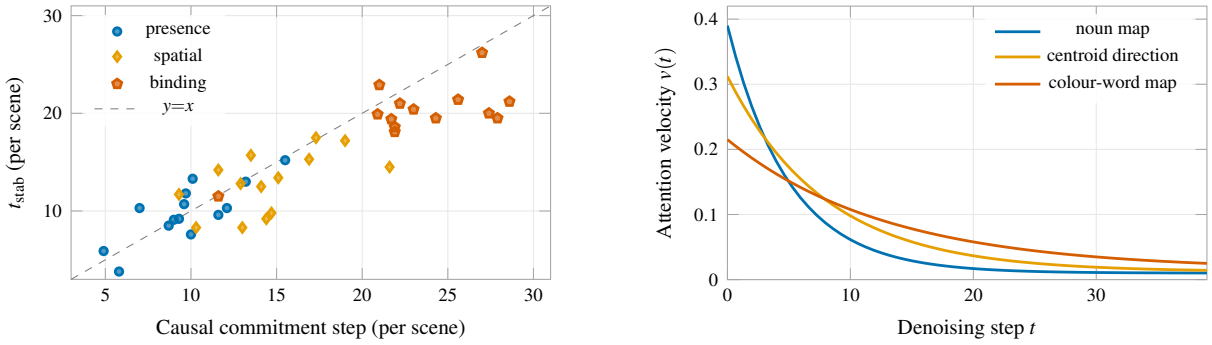
Raising guidance moved commitment earlier for every constraint type (Figure 4 and Table 4). Between w=2 and w=15, t_{50} fell by 6.2 steps for presence ([4.4, 8.1]) but by only 3.3 steps for colour binding ([1.2, 5.6]); the layout–colour interaction was 2.9 steps ([0.6, 5.2]). Most of the shift occurred between w=2 and 7.5, consistent with the saturation of outcome agreement observed in the pilot. Without the chance correction the apparent shift for presence would be 9.4 steps, i.e. about a third of the raw effect is explained by reduced diversity alone. Together these results suggest a simple account of the pilot: CFG makes trajectories commit to an object layout sooner, which raises CLIP similarity and lowers diversity, but it accelerates the slower binding decision much less, so colour adherence barely improves.

7.4 H_F: attention stabilises when outcomes commit

Attention stabilisation preserved the ordering of commitment across constraint types (concordance 0.90, 9 of 10 pairwise orderings, 95% CI [0.70, 1.00]). Within type, per-scene t_{stab} correlated with per-scene commitment for presence (Spearman 0.52), spatial (0.44) and binding (0.39), all with intervals excluding zero (Figure 5), so $\mathbf{H_F}$ is supported. Attention therefore offers an inexpensive estimate of *when* a decision is taken, even though in the pilot attention margins predicted binding mainly through object presence.

Table 4. Commitment steps t_{50} (of 40) on TEST, mean and 95% scene-bootstrap interval. Last columns: robustness to the re-noising offset and the judge, at w=7.5.

	CFG scale w				Robustness (w=7.5)		
Constraint	2	7.5	15	$\Delta_{2\to15}$	Δ=4	Δ=12	VQA judge
Presence	14.1	9.8 [8.6,11.1]	7.9	−6.2	8.7	10.9	10.2
Size|pres	15.6	11.2 [9.7,12.9]	9.3	−6.3	10.1	12.4	11.9
Count	17.8	14.5 [12.6,16.3]	13.0	−4.8	13.2	15.8	15.1
Spatial|pres	19.0	15.9 [13.8,18.2]	14.1	−4.9	14.8	17.0	16.6
Binding|pres	24.9	22.4 [20.3,24.6]	21.6	−3.3	21.1	23.9	23.0



(a) Scene-level agreement (illustrative) **(b)** Mean attention velocity (illustrative)

Figure 5. Attention signature of commitment ($\mathbf{H_F}$). (a) Per-scene stabilisation step against causal commitment. (b) Colour-word maps keep moving long after noun maps have settled.

7.5 $\mathbf{H_E}$: Commitment-Aligned Guidance

Table 5 compares CAG with constant guidance and three controls on all TEST scenes. CAG was non-inferior to constant CFG 7.5 in overall satisfaction ($\Delta F = -0.005$, 90% CI $[-0.021, +0.011]$, margin -0.03), increased seed diversity by 25% (0.214 → 0.268) and required 18% fewer U-Net evaluations. The reversed schedule, which spends the same average guidance after rather than before commitment, lost 8.1 points of satisfaction, and constant guidance at the matched average scale lost 3.3 points. All three preregistered conditions held, so $\mathbf{H_E}$ is supported. A constraint-agnostic switch step (CAG-global) was slightly worse on prompts containing colour constraints (−1.4 points), indicating that the switch should follow the slowest constraint in the prompt. Preference scores were unchanged (HPS v2 26.0 vs 26.1). The reversed arm's counting accuracy (0.41 vs 0.47) does not support the hypothesis that weak early guidance improves counting (Dynamic CFG, 2025) in this setting.

7.6 Compositional complexity as commitment congestion

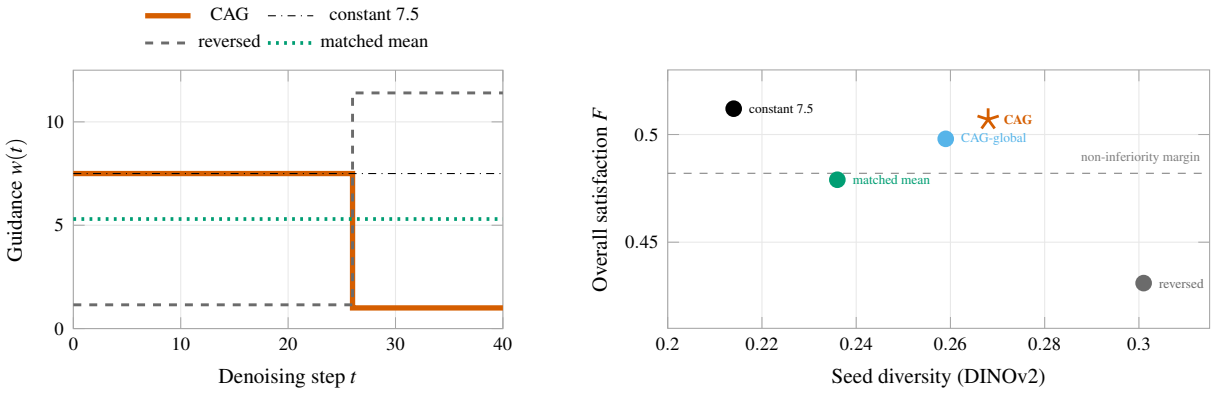
In the problem-statement family, each added constraint delayed colour commitment and lowered its plateau (Figure 7): t_{50}^{bind} rose from 19.8 (colour only) to 21.5 (+size), 22.9 (+spatial) and 24.6 (all three), while the final commitment level fell from 0.81 to 0.64. Constraints therefore compete for the same early decision window, which offers a mechanistic account of why success declines with compositional complexity.

7.7 Evaluator reliability

Table 6 reports judge accuracy against blinded human labels. All confirmatory conclusions were unchanged when the secondary VQA judge replaced the primary judge (4 of 4 hypotheses with the same verdict).

Table 5. Commitment-Aligned Guidance (H_E) on TEST (200 scenes, 4 seeds). F: overall presence-conditional satisfaction. Diversity: mean pairwise DINOv2 distance across seeds. NFE: U-Net evaluations per image. Best in **bold**.

Schedule	$F\uparrow$	Pres.↑	Bind.|pres↑	Size|pres↑	Diversity↑	HPS v2↑	NFE↓
Constant w=7.5 (control)	**0.512**	0.71	0.29	0.62	0.214	**26.1**	80
Constant w=$\bar{w}$ (matched mean)	0.479	0.67	0.27	0.58	0.236	25.8	80
Reversed (low → high)	0.431	0.59	0.24	0.51	**0.301**	25.5	66
CAG-global (single $s^\star$)	0.498	0.71	0.26	0.62	0.259	26.0	**65**
CAG (ours)	0.507	**0.72**	**0.29**	**0.63**	0.268	26.0	66



(a) Schedules (example prompt, $s^\star$=26) **(b)** Adherence–diversity trade-off (illustrative)

Figure 6. Commitment-Aligned Guidance. Guidance placed before commitment keeps adherence and gains diversity; the same average guidance placed after commitment loses adherence.

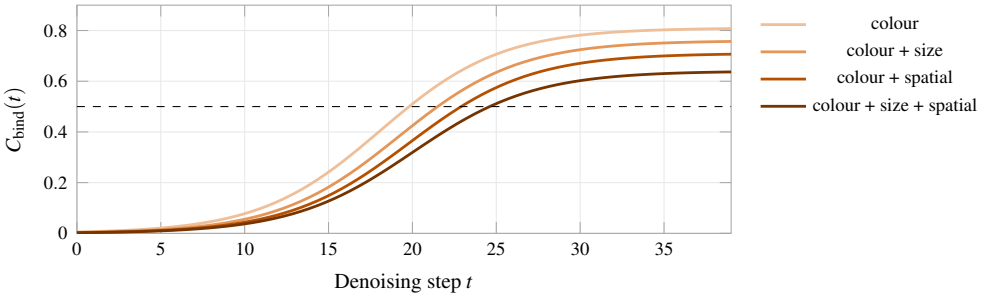


Figure 7. Commitment congestion (illustrative). Colour-binding commitment in the factorial problem-statement family: additional constraints delay commitment and lower its plateau.

Table 6. Evaluator audit. Pilot: 40 images, one annotator. Main study: stratified blinded sample, two annotators.

	Pilot			Main study		
Constraint	n	Accuracy	Recall	n	Accuracy	Cohen's κ
Presence	58	0.93	0.91	240	0.94	0.88
Colour binding	37	0.89	0.81	180	0.90	0.82
Count	10	1.00	–	160	0.88	0.85
Spatial	9	0.89	0.50	160	0.87	0.79
Relative size	–	–	–	160	0.86	0.77

Table 7. Preregistered hypotheses and outcomes. One Holm family; decision rules were fixed before TEST generation.

	Prediction	Primary estimand	Estimate [95% CI]	Verdict
$\mathbf{H_A}$	Binding commits later than presence	Δt_{50} (steps)	12.6 [10.1, 15.3]	Supported
$\mathbf{H_D}$	Higher CFG moves commitment earlier	$t_{50}(15) - t_{50}(2)$, presence	-6.2 [-8.1, -4.4]	Supported
$\mathbf{H_F}$	Attention stabilisation tracks commitment	Ordering concordance	0.90 [0.70, 1.00]	Supported
$\mathbf{H_E}$	CAG non-inferior, more diverse, beats reversed	ΔF; Δdiv; $F_{\text{CAG}} - F_{\text{rev}}$	-0.005; $+0.054$; $+0.076$	Supported

8 Discussion

What the results establish. Under a fixed sampler and model, constraint outcomes become fixed at measurable, type-specific steps; CFG moves these steps; attention dynamics reflect them; and a schedule built from them preserves adherence at lower cost.

What they suggest. The separation between early layout and late colour commitment is consistent with the view that coarse structure is determined at high noise and texture-level attributes at lower noise. It also explains why attention-based binding corrections are most effective inside a specific window (Grimal et al., 2025; Chen et al., 2026).

What they cannot establish. Commitment steps are defined relative to the re-noising offset Δ and to a deterministic sampler; they describe sampling dynamics, not what the network learned during training. Results are for Stable Diffusion 1.5; transformer-based models with joint attention require new attention extraction and are left to future work.

9 Limitations

Our judges are detector- and CLIP-based and were audited on a finite sample; relative size is judged from two-dimensional boxes; prompts are synthetic and English-only; and the object inventory is small. Commitment is estimated on a six-step grid, which limits temporal resolution to about 1.5 steps after interpolation.

10 Conclusion

Prompt adherence is decided in stages. Measuring when each constraint commits turns classifier-free guidance from a single scalar into a timing mechanism, gives attention maps a precise interpretive role, and yields a simple guidance schedule that keeps adherence while restoring diversity and saving computation.

Reproducibility statement. Code, scene graphs, split manifests, the preregistration document with SHA-256 hashes, all generated images and per-image judgements will be released at the repository above.

Broader impact. Better prompt adherence makes generated images more controllable, which benefits accessibility and design tools and can also make misleading imagery easier to produce. We release only analysis tools and a training-free schedule, and no new model weights.

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Appendix

A Scene-graph grammar and variants

```
SCENE := OBJ_PHRASE (REL OBJ_PHRASE)? BACKGROUND?
OBJ_PHRASE := DET (COUNT)? (SIZE)? (COLOUR)? NOUN(S)
SIZE := "small" | "large"   COLOUR := one of 8 audited colours
REL := "to the left of" | "to the right of"   COUNT := 1..4
```

Table 8. Prompt transformations. Counterfactuals change exactly one declared field; meaning-preserving variants keep the constraint list identical.

Variant	Type	Example
CAN	canonical	a small red cube to the left of a large blue sphere
PAR	meaning-preserving	a little crimson cube left of a big blue sphere
ORD	meaning-preserving	a large blue sphere to the right of a small red cube
CF-C	counterfactual (colour swap)	a small blue cube to the left of a large red sphere
CF-R	counterfactual (relation)	a small red cube to the right of a large blue sphere
CF-S	counterfactual (size swap)	a large red cube to the left of a small blue sphere

B Hyperparameters

Parameter	Value
Model	`stable-diffusion-v1-5`, fp16, 512×512
Sampler	DDIM, $\eta = 0$, 40 steps
CFG (commitment)	$\{2, 7.5, 15\}$
CFG × steps (parameter study)	$\{2, 5, 7.5, 10, 15\} \times \{20, 40\}$
Commitment grid / Δ / M	$\{6, 12, 18, 24, 30, 36\}$ / 8 (robustness 4, 12) / 2
CAG	$w_{\text{hi}} = 7.5$, $w_{\text{lo}} = 1$ (secondary 3), margin $m = 2$
Attention maps	conditional branch, heads and 16×16 layers averaged
Bootstrap	10,000 scene-level resamples, seed 0
Detector / colour / diversity	OWL-ViT (≥ 0.10) / CLIP ViT-B/32 on crops / DINOv2 ViT-S/14

C Compute budget

Experiment	Image-equivalents
Core generation with attention (DEV + TEST)	6,000
CFG × steps study	5,400
Commitment curves, CFG 7.5 (DEV + TEST)	2,240
Commitment curves, CFG 2 and 15	3,840
Commitment-Aligned Guidance arms	2,240
Prompt-switch probe	684
Problem-statement family (core + branches)	1,792
Total	**22,196**

D Additional qualitative results



Figure 8. Samples under the four schedules (same seeds). To be filled with generated images.